

Which gas would occupy more volume at a constant temperature and pressure, 1.5 g of N₂ gas or 1.5 g of O₂ gas?

- ✔ ☒ A. N₂ because there are more moles of N₂. [61%]
- ☐ B. O₂ because its molecules are larger. [11%]
- ☐ C. Neither gas occupies any volume. [3%]
- ☐ D. Both gases occupy equal volumes. [22%]

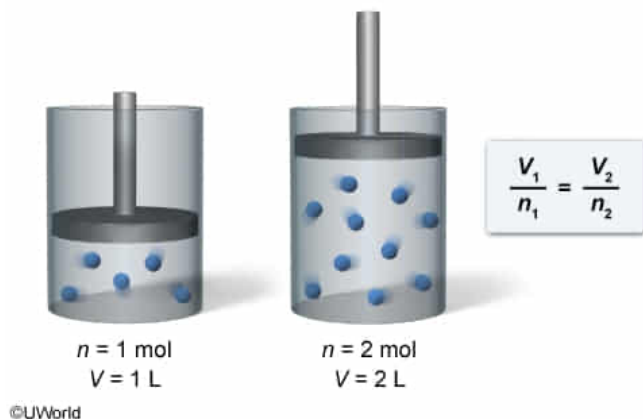
Correct

61% answered correctly

Explanation:

Avogadro's law

The volume occupied by a gas is directly proportional to its number of moles (at constant temperature and pressure).



At a constant temperature and pressure, the volume that a gas occupies depends on its number of moles (or molecules). This is known as **Avogadro's law**, and it can be expressed as the following equation (V is volume and n is number of moles):

$$\frac{V}{n} = \text{constant}$$

In other words, **volume is directly proportional to the number of moles**. Therefore, the gas with more moles occupies a larger volume. When the mass of a gas is given, its number of moles can be determined by dividing by its molar mass (mass per mole):

$$\# \text{ of moles} = \frac{\text{mass}}{\text{mass per mole}}$$

Because the masses of both N_2 and O_2 are equal (1.5 g each), the **gas with the lower molar mass has more molecules**. Using the [periodic table](#), the molar mass of N_2 is calculated to be 28 g/mol, and the molar mass of O_2 is 32 g/mol. Therefore, the **N_2 gas** has more moles and occupies a larger volume.

(Choice B) The size of a gas' molecules is negligible compared to the volume the gas occupies. In addition, oxygen atoms have a *smaller* [atomic radius](#) than nitrogen atoms, which gives O_2 molecules a *smaller* size than N_2 molecules. However, O_2 molecules do have more mass in the nuclei than N_2 molecules, which decreases the number of moles of O_2 in a 1.5-g sample.

(Choice C) Gases assume the size and shape of their container and therefore have the same volume as the container. This is different from size of the molecules themselves, which is considered to be negligible.

(Choice D) The volume occupied by two gases is equal if they share the same pressure, temperature, and number of moles. However, there are more moles of N_2 than moles of O_2 in a 1.5-g sample.

Educational objective:

Avogadro's law states that the volume occupied by a gas is directly proportional to its number of moles if the temperature and pressure are constant. When two gases at the same temperature and pressure are compared, the gas with more moles occupies the larger volume.

Time spent:0 sec

Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws